Terminology

Boot Loader

A small program that loads the operating system

Boot Manager

Lets you choose what OS to boot
especially in systems with multiple operating systems

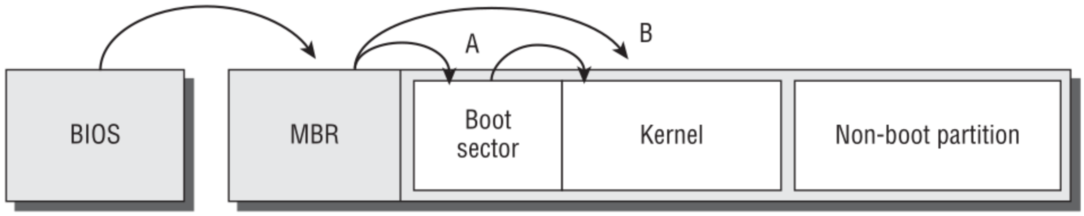
boot manager decides what to boot
boot loader loads the operating system

GRUB

- GRUB acts as a boot manager and a boot loader.
- when you see the menu, it's the boot manager
- when you make a selection, it's a boot loader and loads the selected OS

Boot Loader Principles

BIOS Booting



1. **Firmware** loads **BIOS**
2. in BIOS configure boot order for bootable devices (e.g. hard drive, then USB)
3. from bootable device BIOS loads **Master Boot Record (MBR)**
   MBR contains the **Primary Boot Loader**

Primary Boot Loader:

1. loads **Boot Sector** from device
2. the Boot Sector contains a secondary boot loader which when executed loads the OS kernel

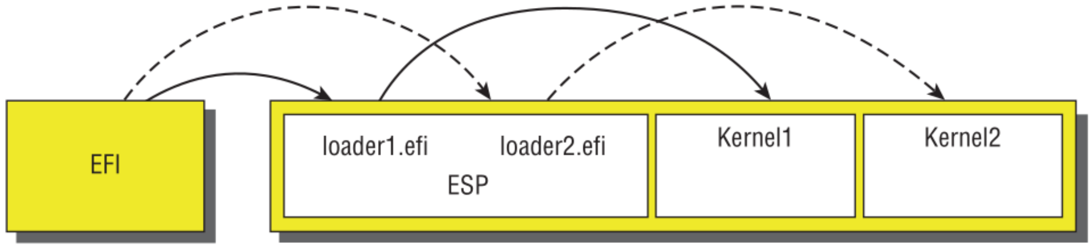
- Windows uses (path A) - boot sector loads Windows Boot Manager
- Linux uses either (path A) or (path B)

EFI Booting

EFI System Partition (ESP)
Boot Loaders are stored as files in a disk partition known as the EFI System Partition (ESP)
ESP is mounted at `/boot/efi`

Boot Loaders
stored in .efi files
e.g. `/boot/efi/EFI/ubuntu/grub.efi`
lets you store a separate boot loader for each OS installed on the computer

Boot Manager
the EFI firmware includes a Boot Manager program
it lets you select which boot loader to launch



GRUB Legacy

Boot Loader for Linux
Eclipsed by GRUB 2

Configuration

- configuration file `/boot/grub/menu.lst`
- GRUBs root partition is the partition in which the configuration file exists
- normally `/boot/grub/`

Options

`default=`	default OS 0 = first OS listed, 1 = second etc
`timeout=`	how long to wait for user input before booting the default OS

Disk Drives

- Legacy GRUB numbers devices
- */dev/hda* becomes (hd0)
- */dev/hdb* becomes (hd1)
- */dev/hda* first partition becomes (hd0,0)
- drive mappings are found in */boot/grub/device.map*

Image Options

refers to the GRUB Boot Loaders executable code

`title`	**Title** label to display when the boot loader runs
`root`	**GRUB Root** location of GRUB root partition
`kernel`	**Kernel Specification** location of Linux Kernel and any kernel options
`initrd`	**Initial RAM Disk** loads drivers into RAM that let the kernel mount the main root filesystem

Adding a Kernel to GRUB

1. load grub.conf in a text editor
2. copy a working configuration for a Linux kernel
3. title → change to a new name
4. kernel → point to new kernel

Reboot and you should see your new kernel in the menu

Installing GRUB

you must specify the boot sector device
`grub-install /dev/sda`
or `grub-install '(hd0)'`

Using GRUB

The first screen of the boot loader shows you a list of operating systems
To edit an entry:

1. select is and press `E` → brings up a new menu
2. highlight the `kernel` line and press `E` to edit kernel options
3. add any options
4. press `enter` to complete edit

GRUB 2

Works with BIOS and EFI (Extensible Firmware Interface)

Configuration

Config file `/boot/grub/grub.conf`
example

```
#
# Kernel Image Options:
#
menuentry "Fedora (3.4.1)" {
   set root=(hd0,1)
   linux /vmlinuz-3.4.1 ro root=/dev/sda5 mem=4096M
   initrd /initrd-3.4.1
}
menuentry "Debian (3.4.2-experimental)" {
   set root=(hd0,1)
   linux (hd0,1)/bzImage-3.4.2-experimental ro root=/dev/sda6
}
#
# Other operating systems
#
menuentry "Windows" {
   set root=(hd0,2)
   chainloader +1
}
```

`menuentry`	title
`set root =`	sets root directory partitions start from 1, not 0

Editing Configuration

Don't edit `/bot/grub/grub.conf`

- instead edit `/etc/grub.d` and `/etc/default/grub`
- then use `update-grub` to makes changes to grub.conf
- when you run update-grub it outputs to standard output. To save the changes `update-grub > /boot/grub/grub.cfg`# Alternative Boot Loaders

The Linux Kernel

- Since version 3.3.0 it has incorporated an EFI boot loader for *x86* and *x86-64* systems
- on EFI computers, this lets the kernel serve as its own boot loader
- eliminates the need for a separate tool like GRUB

Damaged Boot Loader

Linux sometimes becomes un-bootable if the boot loader becomes damaged

1. Try booting from the USB you used to install the OS, and look for an option to boot a kernel from the hard dist
2. once the system is booted, you can use `grub-install` to reinstall GRUB
3. The USB may also provide a recovery option